

Yash Joshi

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EDUCATION

•Master's in Data Science

Jan 2023 - Dec 2025

The University of Texas at Arlington

CGPA: 4.0

Coursework: Big Data Management, Data Science, Data Visualization, Machine Learning, R Programming, Statistics

EXPERIENCE

•Universal Recycle Solutions | AI & Machine Learning Engineer

Feb – Jun 2022

- Designed and deployed **Power BI dashboards** connected to SQL Server and Excel-based logs, allowing management to track KPIs i.e. **waste collection volume, vehicle utilization, route efficiency, turnaround time, fuel usage, and recycling yield**.
- Integrated **predictive analysis modules** using embedded Python scripts, enabling management to forecast waste volume trends by city zone and proactively allocate resources.
- Developed an **ML pipeline** (Random Forest, Gradient Boosting) to forecast collection volume 7–14 days ahead, incorporating weather, seasonality, and historical route data as features.
- Delivered dashboards that combined actual vs predicted KPIs, enabling supervisors to re-route vehicles and reassign drivers, cutting overtime costs by an estimated **12% per quarter**.

•Larsen and Toubro Private LTD. | Data Engineer

Dec 2022 – Feb 2023

- Configured and managed **data ingestion pipelines** connecting **SAP ERP, SQL Server, and flat-file feeds**, enforcing schema validation and ensuring **99.5% data integrity** across finance and operations datasets.
- Engineered **ETL workflows** in Python (Pandas, SQLAlchemy) and **Airflow**, automating ingestion and transformation tasks, which reduced manual reporting preparation time by **35%**.
- Optimized pipeline performance by implementing **parallel batch jobs, query optimization, and indexing strategies**, cutting average data refresh latency from **2 hours to 25 minutes**.
- Developed **data quality monitoring scripts** (schema drift detection, anomaly flagging, and null checks) that lowered data-related production errors by **40%**.

PERSONAL PROJECTS

•Career Prediction System: Web application based on Multi class Classification Algorithm

The tool used ML algorithms to analyze user data and provide personalized career suggestions, addressing suitable job roles.

- Built a multiclass classification model using Random Forest & One-vs-Rest, achieving 92% accuracy in predicting personalized career paths
- Created and published a custom dataset on Kaggle (Skill & Career Recommendation Dataset), achieving 5,594+ views and 1,164+ downloads with a strong 0.20 engagement rate demonstrating real-world relevance and adoption by the global data science community
- Technology Used: Python, Reactjs, Postgres, OnevsRest, Random Forest.

•KSAT Quest: Regression Runoff Prediction

Predicted soil saturated hydraulic conductivity (Ksat) from the UKSAT dataset via a reproducible ML pipeline.

- Delivered measurable performance: R^2 - 0.55 on best configuration, with RMSLE- 0.073 and RMSE- 0.81 on held-out validation robust for hydrological modeling use cases.
- Benchmarked multiple supervised regression algorithms (Ridge, Linear Regression, MLPRegressor, KNN, Decision Tree, Gradient Boosting) with cross-validation (KFold) and hyperparameter tuning (GridSearchCV)

•E-Voting using Blockchain

Designed a system leveraging blockchain technology to ensure immutable and transparent vote recording and fraud

- Provided end-to-end encryption and real-time verification of votes, ensuring voter anonymity and data integrity.
- Integrated Smart Contracts to automate vote validation and tallying processes, reducing manual intervention while enhancing system reliability.
- Technology Used : Solidity Smart contract , Ethereum, Node.js, Ganache, IPFS.

TECHNICAL SKILLS AND INTERESTS

Languages: Python, Sql, Scala, Java, R programming, Shell scripting

Libraries : Pandas, Python Libraries, ReactJs, Numpy, Pyspark

Web Dev Tools: Nodejs, VScode, Git, Github, Tableau, Power Bi

Frameworks: ReactJs, Apache Spark, Flask, Django, Heroku

Cloud/Databases: MongoDB, AWS, Firebase, Relational Database(mysql)

LEADERSHIP & AWARDS

Event	Role	Outcome
Hackathon UTA (Apr 2024)	Team Lead	1st Place
Hack Overflow (Feb 2022)	Team Lead	1st Place
State Level Paper Presentation (Sep 2019)	Presenter	1st Place
Smart India Hackathon (Dec 2022)	Team Leader	Finalist (2nd Place)